Tables	Our Reference Texts	LLM Generated Text

	Test-standard			
	Yes/No	Num	Other	All
Prior (Goyal et al., 2017)	61.20	0.36	1.17	25.98
Language-only (Goyal et al., 2017)	67.01	31.55	27.37	44.26
MCB (Fukui et al., 2016)	78.82	38.28	53.36	62.27
BUTD (Anderson et al., 2017)	82.20	43.90	56.26	65.32
VQA-E (Anderson et al., 2017)	83.22	43.58	56.79	66.31
Beyond Bilinear (Yu et al., 2018)	84.50	45.39	59.01	68.09
ours-single	**84.69**	**46.75**	**59.30**	**68.37**
ours-ensemble-10	**86.15**	**47.41**	**60.41**	**69.66**

Caption: Comparison of our results in VQA task with the state-of-the-art VQA methods on validation set and test-standard set. Accuracies in percentage (%)are reported.

Our system outperforms the state-of-the-art systems which indicates the effectiveness of including caption features as additional inputs. **In particular, we observe that our single model outperform other methods especially in 'Num' and 'Other' categories.** The reasons are that the additional captions are capable of providing more numerical clues for infering 'Num' type questions and more common knowledge for answering the 'Other' type questions. Furthermore, an ensemble of 10 single models results in 69.7% in the test-standard split.

This table compares the performance of different methods on a visual question answering (VQA) task. It shows the accuracy of each method in percentage, with our method (both single and ensemble versions) achieving the highest accuracy across all categories. The categories include yes/no questions, numerical questions, and other types of questions, as well as an overall score.

mAP	-near	-midrange	-far	-large	-medium	-small
CLIP	19.0	11.9	8.2	3.9	8.2	18.4
Grounding DINO	28.0	26.5	23.5	11.5	32.9	39.0

Caption: Results of image-level and object-level models.

Compared with our adopted object-level grounding DINO, we notice that object-level alignment shows much better results from either a distance perspective or a size perspective. For example, 28.0 AP of near objects is achieved in object-level alignment, **9 points higher than image-level alignment.** We analyze that CLIP requires more accurate bounding box prediction, in comparison with object-level alignment which only requires a nearest box.

For the distance ranges, CLIP has mAP values of 19.0 for near, 11.9 for midrange, and 8.2 for far. Grounding DINO performs better with mAP values of 28.0 for near, 26.5 for midrange, and 23.5 for far. For object sizes, CLIP has mAP values of 3.9 for large, 8.2 for medium, and 18.4 for small. Grounding DINO again performs better with mAP values of 11.5 for large, 32.9 for medium, and 39.0 for small.